1. Are all these models for Short-Term Load Forecasting (STLF)?

Yes, absolutely.

In power engineering and IEEE standards, forecasting horizons are defined as:

- **Ultra-Short-Term:** Seconds to < 1 hour (frequency regulation, battery control).
- **Short-Term (STLF): 1 hour to 1 week (typically day-ahead: 24 to 48 hours).**
- **Medium/Long-Term:** 1 month to several years (grid planning, power plant construction).

Every model in our list is specifically evaluated on **Short-Term horizons**:

- **TCN-QRNN (*Nature 2025*):** Forecasts **24 hours ahead** using past 144 hours.
- **EMSformer (*IEEE TPWRS late 2025*):** Forecasts **24 hours ahead (day-ahead)** at 15-min, 30-min, and 1-hour intervals.
- **GCML (*MDPI Nov 2025*):** Forecasts **12-step and 24-step ahead** (12h to 24h).
- **LTE / LT Hybrid (*IEEE TSG May 2025*):** Forecasts 1-step (15 min) and sequence multi-step (4, 8, 12 steps = 1h to 3h ahead).
- **DLinear, PatchTST, TimeMixer, iTransformer:** Evaluated from short horizons ($T \in \{12, 24, 48, 96\}$) up to longer horizons on electricity datasets.

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2. What Features and Parameters Are Used in Each Paper?

Here is the exact technical parameter and feature comparison:

Model / Paper	Features Used (*X* and *y*)	Lookback Window (*L*)	Prediction Horizon (*T*)	Key Model Hyperparameters
TCN-QRNN *(Nature 2025)*	***y*:** Power consumption (`kWatt/hh`) ***X*:** Temperature, Humidity, Pressure	**144 hours** (288 steps for 30m)	**24 hours** (48 steps)	• TCN Dilations: [1, 1, 3, 3, 5, 5, 8], kernel 3 • QRNN Hidden size: 128 units • Activation: SiLU • Dense head: 64 → 24
LTE / LT Hybrid *(IEEE TSG 2025)*	***y*:** Total cluster power (`kW`) ***X*:** Sub-metered appliance load series (intra-house clusters)	**48 hours** (192 steps for 15m)	**1 step (15m)** or 4, 8, 12 steps	• LSTM Front-end: K=2 layers, L=32 cells • Embedding Dimension: T=256 • Transformer Encoder: P=6 layers, R=4 heads • FFN Dimension: 1024
EMSformer *(IEEE TPWRS 2025)*	***y*:** Active load curve ***X*:** NWP weather (temp, humidity, solar, wind), electricity prices	**72 to 96 steps**	**24 to 48 steps** (Day-Ahead)	• Patch 1: Length N_1=4, Stride S_1=1 • Patch 2: Length N_2=8, Stride S_2=2 • Unmasked ProbSparse Attention • BiLSTM residual output layer
GCML *(MDPI Nov 2025)*	***y*:** Hourly consumption per cluster ***X*:** Clustered load sequences (Autoencoder + K-Means)	**24 hours**	**12 and 24 hours**	• 1D Inception Convolution (kernels 1, 3, 5) • 2D Filter-Attention window size = 5 • Dynamic task loss weighting ($\alpha_d(t)$) • Dropout: 0.1
DLinear *(AAAI 2023)*	***y*:** Target load series ***X*:** Trend stream (X_{Trend}) + Seasonal stream (X_{Seasonal})	**96 to 336 steps**	**24 to 96 steps**	• Moving average kernel size: 25 • 2 single linear layers (`Linear_Trend`, `Linear_Seasonal`) • No attention, no hidden recurrent units
PatchTST *(ICLR 2023)*	***y*:** Target series ***X*:** Channel-Independent univariate patches	**336 to 512 steps** (42 or 64 patches)	**24 to 96 steps**	• Patch length P=16, Stride S=8 • 3 Transformer encoder layers, H=16 heads • Latent dimension D=128, FFN=256 • RevIN Instance Normalization
iTransformer *(ICLR 2024)*	***y*:** Target series ***X*:** Multivariate variables as tokens	**96 steps**	**24 to 96 steps**	• L=2 to 4 Transformer blocks • Variate embedding $D \in \{256, 512\}$ • Inverted self-attention across variables • LayerNorm across temporal series
LightGBM *(Enhanced)*	***y*:** `grid_load_kWh` ***X*:** Our 21 engineered features	**Tabular Lags** (t-1, t-2, t-48, t-336)	**1 step to 48 steps**	• Leaf-wise gradient boosting • Learning rate: 0.05, max_depth: 7 to 9 • 500 to 1,000 boosted trees

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3. How Are the Models Trained and Verified?

Across these papers, training follows a strict **3-step validation protocol**:

```
[ Full Historical Dataset ]
  │
  ├──→ 1. Training Set (70% - 80%): Used by Optimizer (Adam/SGD) to update weights
  │
  ├──→ 2. Validation Set (10% - 15%): Used during training for Early Stopping
  │
  └──→ 3. Test Set (10% - 20%): STRICTLY UNSEEN. Used only for the final error table
```

1. **Chronological Splitting:** Because time series has a strict time arrow, **data is never randomly shuffled**. The test set is always the most recent chronological period (e.g., our test set is Oct 2013 to Feb 2014).
2. **Data Normalization:**
   - Z-score standardization ($x' = (x - \mu) / \sigma$) or Min-Max scaling (0 to 1).
   - **Critical Rule:** The mean (μ) and standard deviation (σ) are calculated **only on the training set**, then applied to the validation and test sets to prevent data leakage.
3. **Optimization & Early Stopping:**
   - Most neural networks use the **Adam optimizer** with initial learning rate 10^{-3} to 5×10^{-4}.
   - If the validation loss does not improve for 2 to 5 consecutive epochs (the "patience" parameter), training stops early to avoid overfitting.

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4. What Are the Performance and Error Metrics Used?

All papers evaluate models using deterministic point error metrics:

Metric	Full Name	Mathematical Formula	What It Measures
MAE	Mean Absolute Error	$\frac{1}{n}\sum\lvert y_t - \hat{y}_t\rvert$	Average error in actual kilowatt-hours (intuitive to utilities).
RMSE	Root Mean Squared Error	$\sqrt{\frac{1}{n}\sum(y_t - \hat{y}_t)^2}$	Penalizes large forecasting blunders during peak load hours.
MAPE	Mean Absolute Percentage Error	$\frac{100\%}{n}\sum\left\lvert\frac{y_t - \hat{y}_t}{y_t}\right\rvert$	Percentage error relative to true load.
WAPE	Weighted Absolute Percentage Error	$\frac{\sum\lvert y_t - \hat{y}_t\rvert}{\sum y_t} \times 100\%$	**Best for smart meters.** Unlike MAPE, it does not explode to infinity when nighttime load drops near zero.
R^2	Coefficient of Determination	$1 - \frac{\sum(y_t - \hat{y}_t)^2}{\sum(y_t - \bar{y})^2}$	Goodness of fit (0 to 1). Values > 0.90 mean the model captures >90% of grid variance.
RSE	Relative Squared Error	Ratio of MSE to ground-truth variance	Used in *EMSformer (IEEE 2025)* to normalize error across different datasets.

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5. How Many Use Probabilistic Forecasting?

- In our list of papers:
  - **Only 1 model** explicitly implements probabilistic forecasting: **EMSformer (*IEEE TPWRS 2025*)**.
  - The **Survey paper (*Zhao et al., IEEE 2026*)** reviews probabilistic loss formulations (Pinball Loss, Quantile Regression, Negative Log-Likelihood).
  - The other models (**DLinear, PatchTST, TimeMixer, iTransformer, TCN-QRNN, LTE, GCML**) focus purely on **deterministic point forecasting**.

6. Is Probabilistic Forecasting Better Than Point Forecasting?

Neither is strictly "better"—they solve **two different operational problems**:

```
Point Forecasting (Single Number):
  "Tomorrow at 6:00 PM, London will consume exactly 742.5 kWh."
  ──→ Used for: Electricity market bidding, generator scheduling, baseline billing.

Probabilistic Forecasting (Range with Confidence Intervals):
  "Tomorrow at 6:00 PM, London will consume 742.5 kWh, with a 90% probability
   that it will fall between 710.0 kWh and 780.0 kWh."
  ──→ Used for: Grid spinning reserves, blackout risk management, renewable hedging.
```

Comparison:

Factor	Point Forecasting	Probabilistic Forecasting
Output	Exact estimated value $\hat{y}_t$.	Probability distribution or quantiles (e.g., 5th, 50th, 95th percentiles).
Loss Function	MSE or MAE.	Pinball Loss (Quantile Loss) or Negative Log-Likelihood.
Complexity	Simple, fast, and easy to interpret.	More complex; must balance interval width (PINAW) against coverage (PICP).
Status in Research	**Mandatory.** Every study must establish point forecast accuracy first.	An optional extension for risk analysis.

> **Verdict for your project:** Start with **deterministic point forecasting** (matching DLinear, PatchTST, iTransformer, and TCN-QRNN). You can add quantile prediction intervals as a feature of your novel model if you want to stand out.

7. Do All Papers Use the Same Evaluation Measures and Methods?

They share a common core (MAE and RMSE), but diverge on percentage metrics:

1. **The Universal Standard (Shared by 100% of papers):**
   - **MAE** and **RMSE** are reported by every single paper without exception.
2. **The "Percentage Error" Split:**
   - *Nature 2025* and *Elsevier 2021* use **MAPE**.
   - *Pentsos et al. (IEEE TSG 2025)* rejected raw MAE and MAPE because they mislead on varying cluster sizes; they invented **PMAE** (MAE / Total Demand).
   - *Fang et al. (IEEE TPWRS 2025)* used **RSE** (Relative Squared Error).
   - *Wan et al. (MDPI 2025)* used $\boldsymbol{R^2}$ alongside MAE/RMSE.
3. **The Data Splitting Split:**
   - *Fang et al. (IEEE)* and *Wan et al. (MDPI)* use **8:1:1** or **8:2** train-test splits.
   - *Mochurad (Nature)* and *our pipeline* use **70% Train, 15% Validation, 15% Test**.

The Rule for Your Project:

To make your benchmark universally comparable to all of them, report these **four standard metrics**:

1. **MAE** (matches all papers)
2. **RMSE** (matches all papers)
3. **WAPE / PMAE** (robust percentage error)
4. $\boldsymbol{R^2}$ **Score** (goodness-of-fit, matches MDPI/IEEE)